School of Computer Science & Engineering

Department of Artificial Intelligence & Machine Learning

B.Tech III Year, Vth Semester (CSE)

COURSEPACK

Machine Learning

2026-2027

<QRCODE>

The Operational Blueprint of the Course

Outcome-Based-Active Learning-Your Guide to the Course

Course Title	Machine Learning										
Course Code	R1UC525B	*Core*					Course Type		***Blended***		
Course Credits	Total	4	Theory	3	Tutorial	0	Practical	1	Project	0	
Pedagogical Philosophy *(within 50 words)*	Student-centered active learning through problem-solving, hands-on programming, collaborative projects, and case studies enables learners to understand, implement, evaluate, and apply machine learning algorithms on real-world datasets while fostering critical thinking, teamwork, ethical AI practices, and lifelong learning.										
Semester/Batch/ Section	Batch	2024-2027	Semester	Vth	Section						
Course Lead	Name: Dr. RAJ KUMAR PARIDA and Dr. ANIL SHARMA				Mobile	9338562836 9999348890	Cabin *(Block, Seat#)*	AI-L08(33) AI- 201 (17)			
Course Facilitator(s)	Name: Dr. Trapti Shrivastava and Mr. Abhishek Gaur				Mobile	8109115573 9554059894	Cabin *(Block, Seat#)*	AI-201(34) AI- 501 ()			

INTRODUCTION

Machine Learning (ML) gives computers the ability to learn without being explicitly programmed. Machine learning is a subfield of artificial intelligence (AI). The goal of machine learning generally is to understand the structure of data and fit that data into models that can be understood and utilized by people. Machine learning allow for computers to train on data inputs and use statistical analysis in order to output values that fall within a specific range. Machine learning facilitates computers in building models from sample data in order to automate decision-making processes based on data inputs.

A. COURSE DESCRIPTION

Course Objective

This course enables students to understand the fundamental concepts, algorithms, and applications of Machine Learning. It develops skills in data preprocessing, model building, evaluation, and deployment, empowering students to apply machine learning techniques to solve real-world problems using modern tools and programming frameworks. The goal of machine learning generally is to understand the structure of data and fit that data into models that can be understood and utilized by people.

Course Content *(Full Syllabus)*

UNIT-1: Basic concepts: Definition of learning systems, Goals and applications of machine learning. Aspects of developing a learning system: training data, concept representation, function approximation. History of ML, Life Cycle of ML, Performance metrices, Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm), Issues in Machine Learning, Data Science Vs Machine Learning, Deep Learning vs Machine Learning. Learning Types- Supervised learning, unsupervised learning and reinforcement learning. Overview of classification: setup, training, test, validation dataset, over fitting.

UNIT-2: Supervised Learning- Regression and Classification: Linear Regression and, Logistic Regression, Perceptron Exponential family, Likelihood estimation, maximum likelihood, Ridge and Lasso Regression, Naïve Bayes Classifier Support vector machines; Model Evaluation and Selection (bias-variance tradeoff); Cross-Validation; Classification Accuracy; Decision tree learning algorithm, Inductive bias, Inductive inference with decision trees, Entropy and information theory, Information gain, Attribute Selection Measures and Tree Pruning, ID-3 Algorithm, Issues in Decision tree learning Random Forest Classifier.

UNIT-3: Unsupervised Learning– Clustering algorithms: K-means (objective function, centroid initialization, convergence criteria, elbow method for K selection, limitations), hierarchical clustering (agglomerative and divisive methods, linkage criteria: single, complete, average, dendrogram interpretation), DBSCAN, EM algorithm (expectation-maximization steps, mixture of Gaussians, latent variable models, convergence, advantages over K-means), clustering evaluation methods, applications and limitations of unsupervised learning.

UNIT-4: Reinforcement Learning: Introduction to Reinforcement Learning, Learning Task, Example of Reinforcemen Learning in Practice, Backpropagation algorithm. Introduction to Q learning.

UNIT-5: Evolutionary algorithm: Introduction to evolutionary algorithms, Genetic algorithm, Components, GA cycle o reproduction, Crossover, Mutation, Models of Evolution and Learning,

Teaching Scheme (*Hours and Credits):*

Course Code	Course Title	In hours per semester					Total Credits (C)* (Total Hours/30)
		Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW)and Self-Learning (SL) (TW+ SL)	Total	
		L	T	P	SL		
R1UC525B	**Machine Learning**	42		28	50	120	120/30=4

*This is as per the new National Credit Framework, which accounts for 30 hrs. of learning as equivalent to 1credit. Those universities that are still following the LTP will transform them into no. of hours and fill in the above table.

Legend:

CI: Classroom Instruction (Includes different instructional/implementation strategies, i.e. Lecture(L), Tutorial(T), Case method, Demonstrations, Video demonstration, Problem-based learning, etc., to deliver theoretical concepts)

LI: LaboratoryInstruction(Includesexperiments/practicalperformances/problem-basedexperiencesinlaboratory, workshop, field or other locations using different instructional/Implementation strategies)

TW: Term work-include assignments, seminars, micro projects, industrial visits, any other student activities etc.

SL: Self-learning, MOOCs, spoken tutorials, online educational resources, etc., if provided in a curriculum structure.

B. OUTCOMES & OBE FRAMEWORK (*as per NHEQF)*

Discipline-wise **Programme Outcomes(POs):**

PO	By the End of This Programme, students will be able to
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyse complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	Modern Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for: independent and life-long learning adaptability to new and emerging technologies and critical thinking in the broadest context of technological change. (WK8).

B.1.A Programme-Specific Outcomes (PSOs):

PSO	Program Specific Outcomes (PSOs) that are unique to the given program
PSO1	Have the ability to work with emerging technologies in Computer Science and Engineering requisite to Industry 4.0.
PSO2	Demonstrate Engineering Practice learned through industry internship and research project to solve live problems in various domains.

*Program Specific Outcomes (PSOs) are unique to the given program and reflect its distinct disciplinary and professional requirements

Course Outcomes (COs):

These are the specific aspects you will be able to do by the end of this course-clear, measurable and the basis on which your learning is assessed.

CO No.	By the end of this course, you will be able to…	*KL(L1-L6)
CO1	Explain the fundamental concepts of machine learning including learning systems, types, applications and performance metrices.	L2

CO2	Apply regression and classification techniques to analyze datasets and develop predictive models for practical machine learning applications.	L3
CO3	Implement unsupervised learning techniques using clustering algorithms to group similar data instances from real-world datasets.	L3
CO4	Evaluate and compare machine learning models using appropriate validation techniques and performance metrics to recommend the most suitable model for real-world predictive applications.	L5
CO5	Design and develop machine learning-based applications incorporating reinforcement learning and evolutionary algorithms	L6

*KL-Knowledge Level

Course Articulation Matrix (CO-PO/PSO):

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	1	-	-	1	-	-	-	-	-	2	2	-
CO2	3	3	2	1	3	-	-	1	1	-	2	3	2
CO3	2	3	2	2	3	-	-	-	-	-	2	3	2
CO4	2	3	-	3	3	-	1	1	-	-	2	2	3
CO5	3	3	3	3	3	1	1	2	1	1	3	3	3

C. COURSE DELIVERY

Delivery Model: Session I (Theory + Lab) and Session II (Theory + Lab)

This course runs in two phases across the semester: Session I, before the Mid-Term Examination, and Session II, after it. Even though it's split this way, consider it as one continuous course, not two separate halves: your learning builds steadily from the first week to the last. The two-phase structure simply helps organise how you learn, check on, and are supported along the way. It doesn't break the course into stand-alone pieces, and your outcomes aren't judged topic by topic or module by module. In practice, you'll cover about 50% of the syllabus in the first six weeks, well before the MTE, and the rest 50% in Session II in the same way. Every part comes with activities and learning experiences chosen to match what you're meant to achieve and to help you build the skills and confidence that goes with it. The Course Outcomes are judged once you've completed the whole course at the end of Session II. Your final result reflects your overall achievement in the course.

Session-wise Course Content (*Entire Course Content): **The course content (syllabus) is divided into 12 approximately equal weeks.** The first six weeks were organized as Session I, followed by MTE and the remaining six weeks as Session II after MTE. Each week shall incorporate appropriate learning activities and pedagogical interventions aligned with the intended learning outcomes and designed to promote the development of the associated skills and competencies.*)

Session-I:1-6Weeks(6Weeks)

Basic concepts: Definition of learning systems, Goals and applications of machine learning. Aspects of developing a learning system: training data, concept representation, function approximation. History of ML, Life Cycle of ML, Performance metrices, Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm), Issues in Machine Learning, Data Science Vs Machine Learning, Deep Learning vs Machine Learning. Learning Types- Supervised learning, unsupervised learning and reinforcement learning. Overview of classification: setup, training, test, validation dataset, over fitting.

Supervised Learning- Regression and Classification: Linear Regression and , Logistic Regression , Perceptron , Exponential family, Likelihood estimation, maximum likelihood, Ridge and Lasso Regression, Naïve Bayes Classifier, Support vector machines; Model Evaluation and Selection (bias-variance tradeoff); Cross-Validation; Classification Accuracy; Decision tree learning algorithm, Inductive bias, Inductive inference with decision trees.

SessionII:7-12Weeks(6Weeks*)*

Entropy and information theory, Information gain, Attribute Selection Measures and Tree Pruning, ID-3 Algorithm, Issues in Decision tree learning, Random Forest Classifier.

Unsupervised Learning– Clustering algorithms: K-means (objective function, centroid initialization, convergence criteria, elbow method for K selection, limitations), hierarchical clustering (agglomerative and divisive methods, linkage criteria: single, complete, average, dendrogram interpretation), DBSCAN, EM algorithm (expectation-maximization steps, mixture of Gaussians, latent variable models, convergence, advantages over K-means), clustering evaluation methods, applications and limitations of unsupervised learning.

Reinforcement Learning: Introduction to Reinforcement Learning, Learning Task, Example of Reinforcement Learning in Practice, Backpropagation algorithm. Introduction to Q learning.

Evolutionary algorithm: Introduction to evolutionary algorithms, Genetic algorithm, Components, GA cycle of reproduction, Crossover, Mutation, Models of Evolution and Learning.

Weekly Lesson Plan (Theory): Driving Questions

This is your week-by-week plan. Each week opens with a Driving Question you will explore across that week's sessions, and it shows the skill and competency you are building, along with the active-learning activity you will take part in.

Week	Driving Question	Topics	COs	Skill/Competency	Pre-reading Materials link
1	How do machines learn from data, and what makes machine learning different from traditional programming?	Definition of learning systems, Goals and applications of ML. Aspects: training data, concept representation, function approximation. History of ML and ML Life Cycle.	CO1	Understand ML fundamentals, identify learning system components, distinguish ML applications, explain ML life cycle and concept representation.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	***Think–Pair–Share:*** *Students identify everyday AI applications and classify whether they involve machine learning. Groups discuss how data enables learning and present their reasoning.*			
	Tutorial/Lab: Problem Statement	Explore ML tools such as Python (scikit-learn), WEKA, and Orange. Load and visualize structured datasets like Titanic. Generate statistical summaries and plots. Apply appropriate data preprocessing.			
2	How do we choose the right machine learning approach and evaluate whether a model performs well?	Performance metrics overview. Machine Learning Approaches overview (Regression, Classification, Clustering, Reinforcement Learning, Genetic Algorithm), Issues in Machine. Data Science vs ML vs Deep Learning, Types of Learning: Supervised, Unsupervised, Reinforcement.	CO1	Compare supervised, unsupervised and reinforcement learning; distinguish ML, DL and Data Science; understand performance metrics and ML approaches.	
	Activity (Approx.. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	Jigsaw Learning: Each group studies one ML approach (Regression, Classification, Clustering, Reinforcement Learning, Genetic Algorithm) and teaches it to classmates using real-world examples.			
	Tutorial/Lab: Problem Statement	Implement linear regression to predict house prices using the Boston Housing dataset. Visualize regression line and evaluate with RMSE and R^2.			

Week	Driving Question	Topics	COs	Skill/Competency	Pre-reading Materials link
3	How can regression models discover relationships between variables and make accurate predictions?	Classification: setup, training/test split, overfitting. Supervised Learning: regression and classification problems, Simple Linear Regression Multiple Linear Regression	CO2	Prepare datasets, perform train-test split, identify overfitting, implement simple and multiple linear regression.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	**Problem-Based Learning:** Students analyze a house-price prediction case, identify dependent and independent variables, build a regression model, and interpret prediction accuracy.			
	Tutorial/Lab: Problem Statement	Implement Logistic Regression on Titanic dataset to predict survival using Pclass, Age, Sex. Evaluate with confusion matrix and ROC curve.			
4	How can regularization and probabilistic learning improve predictive models?	Logistic Regression. Exponential Family, Likelihood Estimation, Maximum Likelihood. Ridge and Lasso Regression.	CO2	Implement Logistic Regression, understand likelihood estimation, maximum likelihood, Ridge and Lasso Regression, interpret classification problems.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	Case-Based Learning: Students compare Logistic, Ridge and Lasso Regression on a disease prediction dataset and discuss overfitting and feature selection.			
	Tutorial/Lab: Problem Statement	Apply Ridge and Lasso on Boston Housing dataset. Compare regression coefficients, MSE, and regularization effects.			
5	Why is Naïve Bayes effective despite assuming feature independence?	Naive Bayes. Naive Bayes Classification. Naïve Bayes Classifier.	CO2	Understand probabilistic classification, apply Bayes theorem, implement Naïve Bayes classifier, interpret prediction probabilities.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	**Case-Based Learning:** Students classify spam emails or sentiment data using Naïve Bayes and discuss how probability influences classification decisions.			
	Tutorial/Lab: Problem Statement	Apply Naive Bayes classification to the Adult Census dataset to predict target variable based on Bayes Theorem.			

6	How do Support Vector Machines identify optimal decision boundaries for complex classification problems?	Linear Discriminant Analysis Support Vector Machines (SVM) Bias-Variance Tradeoff	CO2	Understand Linear Discriminant Analysis, SVM, kernel concepts, margin maximization, bias-variance tradeoff, model generalization.	
	Activity (Approx.. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	Problem-Based Learning: Student groups compare SVM and LDA for classifying the same dataset and justify which algorithm performs better under different scenarios.			
	Tutorial/Lab: Problem Statement	Apply Support Vector Machine (SVM) for multi-class classification on the IRIS dataset, experiment with different kernels, Evaluate model and visualize decision boundaries.			

7	How do decision trees identify the best features for making accurate predictions, and why do ensemble methods improve performance?	Entropy and Information Theory, Information Gain, Attribute Selection Measures, Tree Pruning, ID3 Algorithm	CO2	Understand entropy, information gain, attribute selection, tree pruning, ID3 algorithm, decision tree construction, and Random Forest classification. Develop analytical skills for selecting suitable classification models.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	Problem-Based Learning: Students are given a small dataset and manually compute entropy and information gain to identify the root node before constructing a decision tree.			
	Tutorial/Lab: Problem Statement	Train Decision Tree and Random Forest on Titanic dataset to classify survival. Visualize tree and feature importance. Perform pruning and analyze accuracy. Use Logistic Regression and Decision Tree with 10-fold CV on Titanic dataset. Plot learning curves and discuss bias-variance tradeoff.			
8	Why do ensemble methods such as Random Forest often outperform a single decision tree, and how can unlabeled data be grouped effectively?	Advanced Decision Tree Models: Issues in Decision Tree Learning, Random Forest Classifier. Introduction to Clustering: Overview of Unsupervised Learning, K-Means Clustering, Objective Function, Centroid Initialization	CO2, CO3	Understand the limitations of decision trees, Random Forest classification, clustering concepts, objective function of K-Means, centroid initialization, and clustering applications.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	Case-Based Learning: Students compare Decision Tree and Random Forest for disease prediction and discuss situations where clustering is more appropriate than classification.			
	Tutorial/Lab: Problem Statement	Cluster customers based on their Annual Income and Spending Score using the K-Means algorithm. Use the Elbow Method to find the optimal number of clusters and visualize the results.			
9	A larger question to be explored in the week (3 sessions).	Flex Week: Faculty may utilize this flex week to compensate for missed classes or laboratory classes during Session II by providing remedial teaching, laboratory work, assignments, assessments, and other academic support activities.			
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)				
	Tutorial/Lab: Problem Statement				
MTE					
10	How can the optimal number of clusters be determined, and how do different clustering techniques identify hidden patterns?	K-Means Clustering: Convergence Criteria, Elbow Method for Optimal K Selection, Limitations of K-Means. Hierarchical Clustering: Agglomerative and Divisive Clustering, Linkage Criteria (Single, Complete, Average), Dendrogram Interpretation	CO3	Implement K-Means, determine optimal K using the Elbow Method, perform Hierarchical Clustering, interpret dendrograms, and compare clustering techniques.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	**Jigsaw Learning:** Groups become experts in K-Means or Hierarchical Clustering and explain the algorithm, advantages, limitations, and real-world applications to peers.			

	Tutorial/Lab: Problem Statement	Group customers by income and spending behavior using agglomerative clustering. Create and interpret dendrograms.

Week	Driving Question	Topics	COs	Skill/Competency	Pre-reading Materials link
11	Which clustering algorithm is most suitable for datasets with arbitrary shapes, noise, and complex distributions?	Advanced Clustering: DBSCAN, Expectation-Maximization (EM) Algorithm, Mixture of Gaussians, Latent Variable Models, Convergence, Advantages over K-Means, Clustering Evaluation Methods, Applications and Limitations of Unsupervised Learning	CO3, CO4	Understand DBSCAN, Expectation-Maximization (EM), Gaussian Mixture Models, clustering evaluation methods, and performance comparison of clustering algorithms.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	Case-Based Learning: Students analyze datasets with noisy and non-spherical clusters and recommend the most suitable clustering algorithm with justification.			
	Tutorial/Lab: Problem Statement	Apply DBSCAN and EM to structured bank marketing dataset. Compare silhouette score and visual clustering patterns.			
12	How can an intelligent agent learn optimal decisions through interaction with its environment?	Reinforcement Learning: Introduction to Reinforcement Learning, Learning Task, Reinforcement Learning in Practice, Backpropagation Algorithm, Introduction to Q-Learning	CO5	Understand reinforcement learning concepts including agent, environment, states, actions, rewards, policy, value function, Q-learning, and reinforcement learning applications.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	Simulation-Based Learning: Students model a simple navigation problem and identify states, actions, rewards, and optimal policies before implementing Q-learning.			
	Tutorial/Lab: Problem Statement	Design a grid world and implement Q-learning to derive optimal policy. Track value updates over episodes.			
13	How can evolutionary principles be applied to solve complex optimization problems that traditional algorithms struggle with?	Evolutionary Algorithms: Introduction to Evolutionary Algorithms, Genetic Algorithm, Components of GA, GA Cycle of Reproduction, Selection, Crossover, Mutation, Models of Evolution and Learning, Course Revision and Integrated Problem Solving	CO5	Understand Genetic Algorithms, chromosome representation, fitness function, selection, crossover, mutation, reproduction cycle, and optimization strategies.	
	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	Problem-Based Learning: Student teams design a Genetic Algorithm to solve an optimization problem and compare its performance with a conventional search technique.			
	Tutorial/Lab: Problem Statement	Create tabular policy using value iteration. Compare policy path with Q-learning results in same environment. Solve the 0/1 Knapsack problem using GA. Implement selection, one-point crossover, and bit-flip mutation. Visualize fitness evolution over generations.			
14	A larger question to be Explored in the week (3 sessions).				

	Activity (approx. 20 mins.), namely, Jigsaw, Case-Based, problem-based, etc.)	***Flex Week:*** *Faculty may utilize this flex week to compensate for missed classes or laboratory classes during Session II by providing remedial teaching, laboratory work, assignments, assessments, and other academic support activities.*
	Tutorial/Lab: Problem Statement	
	ETE	

Complex Problem/Complex Engineering Problem(exercise): *Please list12ComplexProblems/Complex Engineering Problems (CEP)thatwillserveasthetargetlearningexercisesforstudentsthroughoutthiscourse.*

Week	Complex Problem/ Complex Engineering Problem (CEP)	BTL	SOLO Level	CO
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

Self Learning: MOOCs, spoken tutorials, online educational resources, surveys, clinicals, focus group studies, seminars, etc., if provided in a curriculum structure: two to three SWAYAM/NPTEL/MOOC certifications that complement the course.

S.No	Recommended Course / Learning Resources	Credits	Link	BTL	CO
1					

D. ASSESSMENT&EVALUATION

Assessment Pattern for Theory Course:

Type of Course (T)	CIE			Total Marks		Final Marks CIE*0.5+SEE*0.5
	Lab Work+ Record@	MTE	Lab Exam*	CIE	SEE	
THEORY(Blended)	25	50	25	100	100	100

@ Lab work- 15 marks +Lab record-10 marks

*passing criteria-30% of marks to be secured in the lab exam exam conducted by two examiners (one internal, one external)

E. ACADEMIC REQUIREMENTS

1. **Attendance Requirement**: You must maintain a minimum of 75% attendance in each course, covering all scheduled classes, tutorials, and laboratory sessions.
2. **Class Participation**: You are expected to attend all classes punctually and come well-prepared. Active participation in class discussions, group activities, and practical exercises is essential and contributes to continuous assessment and evaluation.
3. **Practical and Skill-Based Learning**: Attendance in all practical, laboratory, and skill-based sessions is mandatory. You are required to complete all assigned experiments, practical work, and laboratory exercises as prescribed for the course.
4. **Continuous Assessment**: You must appear for all scheduled assessments and Mid-Term Examinations (MTEs). Performance in these assessments contributes significantly to the award of internal marks and overall academic achievement. Absence from such assessments may adversely affect course performance and the cumulative Grade Point Average (GPA).
5. **Plagiarism Policy**: Use of unfair means in any of the evaluation components will be dealt with strictly, and the case will be reported to the examination committee.

F. LEARNINGRESOURCES

F1.Bibliography

- **Textbook:**
  - Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
  - Ethem Alpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press 2004.
- **Reference Books:**
  - Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
  - Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.

F2.Learning Resources

- **Journals/Reports/Industry Trends:**
  - IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
  - Journal of Machine Learning Research (JMLR)
  - Gartner Magic Quadrant for Data Science and Machine Learning Platforms
- **Webliography:**
  - https://github.com/topics/machinelearning
  - https://huggingface.co/huggingface-course
  - https://www.kaggle.com/datasets
- **SWAYAM/NPTEL/MOOC Certifications**:
  - https://nptel.ac.in/courses/106106139
  - https://openlearninglibrary.mit.edu/courses/course-v1:MITx+6.036+1T2019/course/

G. How Your Achievement Is Measured

At the end of the course, your classroom participation, course work and final exams are combined to assess how well you have achieved each Course Outcome. Adopting this entire flow will result in good academic performance.

Prepared by

Dr. Raj Kumar Parida (Course Lead- SoAI)

Dr. Anil Sharma (Course Lead- SCSE)

Verified by _____(ProgramChair)

Verified by _____(HOD)

Semester Preparedness Meeting held on: 24.07.26

Approved By (Dean)

Annexure: A Faculty Guide to Pedagogical Philosophies

Selectinganddeclaringthepedagogicalapproachthatunderpinsyourcourse

1. Purpose of this Annexure

ThisannexureaccompaniestheUniversityCoursePack.IntheCoursePack,everyfacultymemberis askedtodeclarethe pedagogicalphilosophythatunderpinstheircourse.Thishandoutexiststomakethatdeclarationaninformedchoicerather than a routine entry. It explains, in brief, the principal pedagogical philosophies used in contemporary higher education, maps their defining attributes on a single comparative table, and offers illustrative courses drawn from across our multidisciplinary schools, for which each philosophy is well suited.

The intent is not to rank the philosophies. No approach is superior in the abstract. Each is powerful when matched to the rightlearningoutcome, learner and stage ofstudy,and weak when misapplied.Asinglecourse willoftendrawonseveral of them.

Table1: AttributesofDifferentPedagogicalPerspectivesandApproaches

Perspective/ Approach	**Learning seen as** Ext–Int	**Taskdefiner** Stu–Tea	**Problem structure** Ill–Well	**Activity director** Stu–Tea	**Interpersonal interaction** Ind–Soc	**Product output**None –Prod
Behaviourism	X---------	--------- X	- --------X	------- X--	--X--------	-X---------
Cognitivism	----X------	-------- X-	- -------X-	------- X--	--X--------	--X--------
Constructivism	----X------	-----X-----	------X----	----X------	----X------	---X-------
Social constructivism	---X-------	-----X-----	-----X-----	---X-------	- ------X--	-----X-----
Connectivism	-X---------	--X--------	-X---------	--X--------	- ------X--	------X----
Collaborative learning	-----X-----	------X----	-----X-----	---X-------	- ------X--	------X----
Problem-based learning	----X------	------X----	--X--------	--X--------	-------X---	----X------
Inquiry-based learning	-----X-----	---X-------	--X--------	--X--------	-------X---	---X-------
Constructionist learning	-----X-----	----X------	--X--------	--X--------	-------X---	- --------X
Design-based learning	-----X-----	----X------	--X--------	---X-------	-------X---	- --------X
Games-based learning	------X----	-------- X-	-------X---	--X--------	-----X-----	-X---------

HowtodecodeTable1(AttributesofDifferentPedagogicalPerspectivesandApproaches)

Attribute (left↔rightpole)	**Whatitasks**
Learningisseen as (External↔Internal)	Isknowledgetreatedasanexternalrealitytransmittedtothelearner,oras somethingthe learner builds internally from experience?
Taskdefiner (Student↔Teacher)	Whodecideswhatistobelearnedordone,theteachersettingthetask,orthestudents framing their own?
Problemstructure (Ill↔ Well)	Areproblemsopen-endedand messywithmanyvalidpaths(ill-structured),orclearly bounded with defined correct answers (well-structured)?
Activitydirector (Student↔Teacher)	Whodrivesthelearningactivitymomenttomoment:ateacher-ledclassorstudent-led work?
Interpersonalinteraction (Individual↔Social)	Islearninglargelysolitary,ordoesitarisethroughinteractionwithpeersandothers?
Productoutput (None↔Product)	Doestheprocessculminateinatangibleartefactordeliverable?

2. The eleven philosophies at a glance

Pedagogical Philosophy	Core Principle	Best Suited For	Example Courses/ Disciplines
1.Behaviourism	Learning occurs through stimulus, response, repetition, andreinforcement.Focuseson observable behaviour and mastery through practice.	Foundational knowledge, memorisation, procedural skills,drills,andautomatic performance.	Mathematics, Programming Fundamentals, Language Grammar, Pharmacology,Accounting,Aviation Checklists, LaboratorySafety, Legal Maxims
2.Cognitivism	Learninginvolves processing, organising, storing, and retrieving information while developingmentalmodelsand schemas.	Conceptual understanding, analyticalthinking,structured knowledge, and cognitive skill development.	DataStructures,EngineeringMechanics, Thermodynamics, Anatomy & Physiology, Financial Accounting, Constitutional Law, Cell Biology
3.Constructivism	Learners actively construct knowledgebyconnectingnew experiences with prior understanding through exploration and reflection.	Inquiry, experimentation, conceptual understanding, reflection,andexperiential learning.	Engineering Design Labs, Physics Experiments,MarketingCases,Clinical Simulation, Architecture Studios, Environmental Science
4. Social Constructivism	Knowledge is co-created throughcollaboration,dialogue, scaffolding, and shared experiences.	Discussion-basedlearning, teamwork, peer learning, communication, and collaborative problem-solving.	Moot Courts, HR & OB Seminars, LiteratureDiscussions,DesignCritiques, Political Science, Tourism Studies
5.Connectivism	Learning occurs through buildingnetworksofpeople, digital resources, and knowledge sources in a connected world.	Technology-richlearning, emerging technologies, lifelong learning, digital literacy, and information management.	AI & Machine Learning, Digital Marketing, Research Methodology, Open-SourceDevelopment,Aviation Industry Trends
6.Collaborative Learning	Studentsworktogethertoward shared goals through cooperation, communication, and mutual accountability.	Teamprojects, interdisciplinary work, peer interaction,andcollaborative decision-making.	EngineeringCapstones,BusinessPlan Competitions, Community Health Projects, Hackathons, Newsroom Projects, Event Management
7.Problem-Based Learning (PBL)	Learningbeginswithauthentic, ill-structured problems that students investigate before formal instruction.	Professional reasoning, decision-making, critical thinking,andself-directed learning.	Clinical Case Studies, Engineering Troubleshooting, Legal OpinionWriting, BusinessConsultingCases,SupplyChain Problems
8.Inquiry-Based Learning	Studentsinvestigatequestions, collect evidence, analyse findings, and develop conclusions independently.	Research, scientific investigation,evidence-based reasoning, and discovery learning.	ScienceResearchLabs,MarketResearch, Data Analytics, Historical Research, Environmental Studies, Agricultural Investigations
9.Constructionist Learning	Learning occurs most effectivelywhenlearnerscreate meaningful,shareableproducts or artefacts.	Making, designing, prototyping,creativity, innovation,andproject development.	Robotics, IoT Projects, App Development,3DPrinting,Product Design, Entrepreneurship MVPs, DocumentaryProduction
10.Design-Based Learning	Studentssolvereal-world, human-centred problems through iterative design thinking,prototyping,and testing.	Innovation, product development,designthinking, user experience, and solution development.	Product Design, UX/UI, Architecture Studios,EngineeringDesignProjects, SocialInnovation,HospitalityService Design
11.Games-Based Learning	Learningisfacilitatedthrough games,simulations,role-play, and gamification with clear goals and feedback.	Skill practice, decision-making, engagement, simulations,andexperiential learning.	Flight Simulators, Stock Market Simulations, Clinical OSCEs, Mock Trials,CybersecurityCapture-the-Flag, Supply Chain Simulations, Language Learning

3. Choosing your philosophy:

The mapping below is a starting point, not a rule. Match the dominant aim of your course to a philosophy, then confirm against the Table and your Course Outcomes.

If your course aims to…	Consider
Buildfluencyinfacts,rulesorfixedprocedures	Behaviourism;Games-basedlearning(forpractice)
Developdeepconceptualunderstandingofabodyof knowledge	Cognitivism
Havestudentsconstructunderstandingthrough direct experience	Constructivism
Developunderstandingthroughdiscussionand teamwork	Socialconstructivism;Collaborativelearning
Cultivateprofessionalreasoningonmessy,real-world problems	Problem-basedlearning
Fosterinvestigation,evidence-gatheringand discovery	Inquiry-basedlearning
Produceatangibleartefact,prototypeorportfolio	Constructionist/Design-basedlearning
Equiplearnerstonavigatefast-changing,networked knowledge	Connectivism